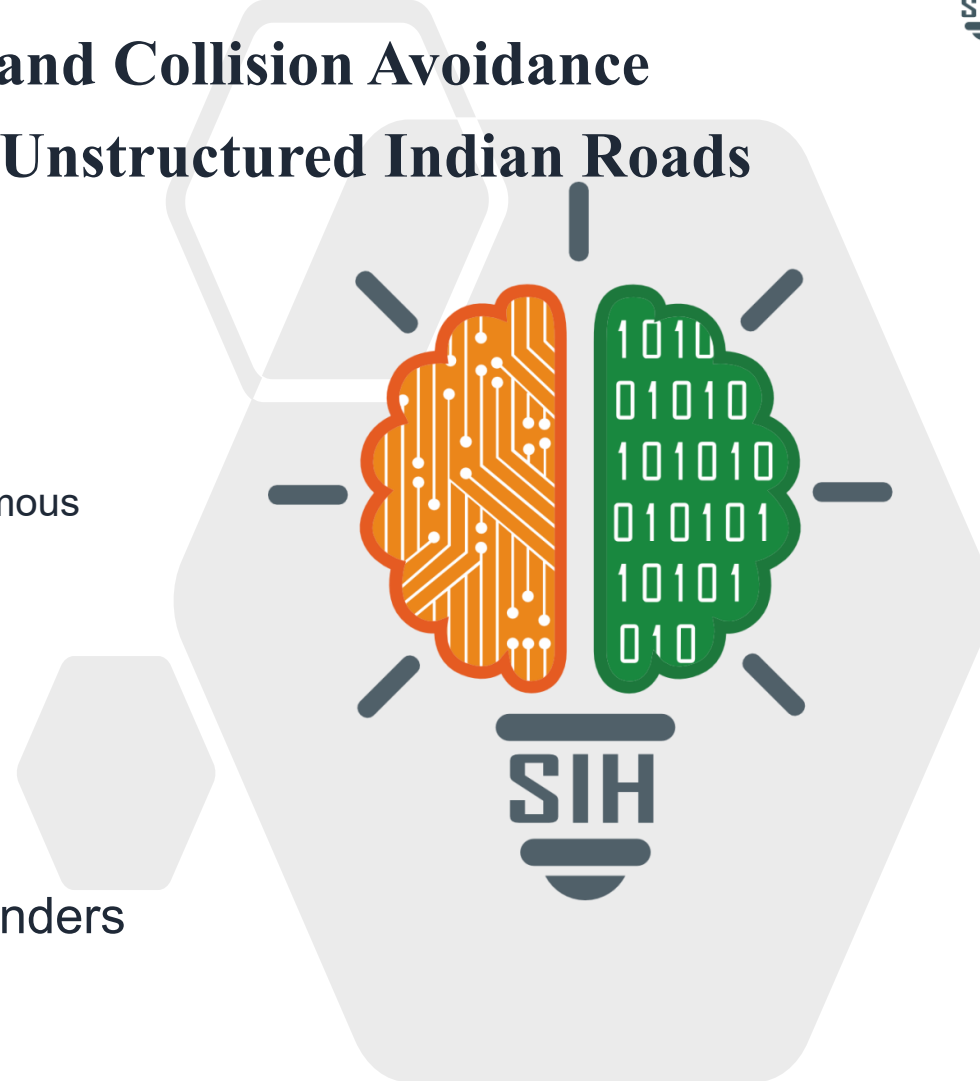


SMART INDIA HACKATHON 2026



Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads

- **Problem Statement ID** – SIH26037
- **Problem Statement Title** –
Adaptive Path Planning and Collision Avoidance for Autonomous
Vehicles on Unstructured Indian Roads
- **Theme** – Smart Vehicles
- **PS Category** – Software
- **Team ID** – 162509
- **Team Name (Registered on portal)** – CraterFinders





PROPOSED SOLUTION



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Problem at Hand

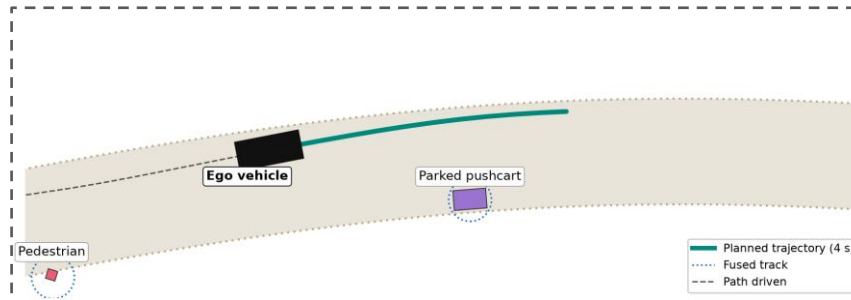
No Lanes to Follow: Village and peri-urban roads are unmarked, so lane-keeping ADAS and AV stacks lose their reference path.

Mixed, Rule-Free Traffic: Autos, cyclists, pedestrians, cattle and pushcarts share one carriageway with no lane discipline.

Unpredictable Intent: A rickshaw may brake, drift or cut in. A single predicted path hides that uncertainty from the planner.

High Human Cost: India recorded 1,68,491 road deaths in 2022 (MoRTH). Autonomy built for Western roads does not transfer.

Our Solution



Closed-loop simulation: in NUDGE mode the ego plans a smooth path past a parked pushcart on an unmarked village road.

Why We Stand Out

- **Lane-free by design:** plans inside a drivable corridor and passes on either side, as Indian drivers do.
- **Working prototype today:** all 5 required scenarios run end-to-end and complete collision-free, scored against ground truth.
- **Zero-licence core:** toolbox-free MATLAB/Octave code that ports directly to Simulink & Stateflow.

Key Features



Corridor Planning

Frenet lattice over free space, not painted lanes.



Multi-Sensor Fusion

Camera + radar + LiDAR into EKF tracks that survive occlusion.



Intent Prediction

Four class-weighted hypotheses per road user.



Behaviour FSM

Cruise, Nudge, Yield, Creep and Stop set the planner's limits.



Safety First

Risk-dominated cost, 0.45 m margin and emergency braking.



Real-Time

10 Hz replanning; 51–66 ms mean cycle.

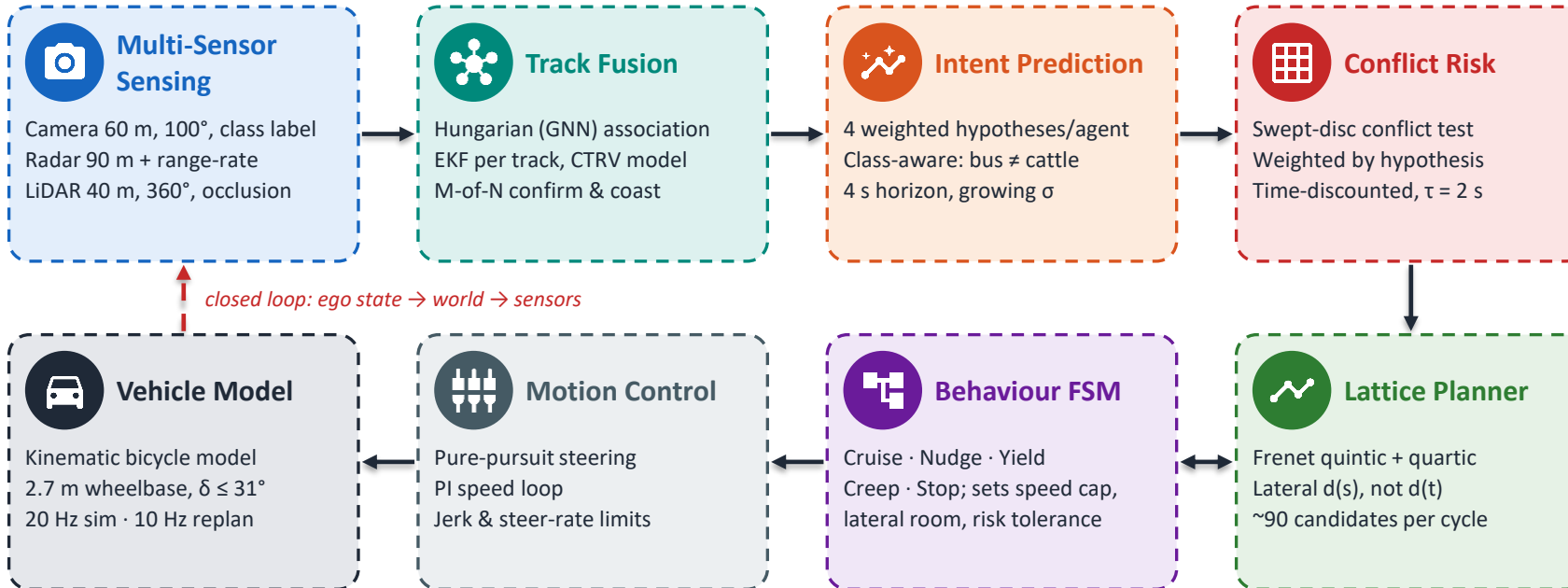


TECHNICAL APPROACH



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Closed-Loop Autonomy Pipeline

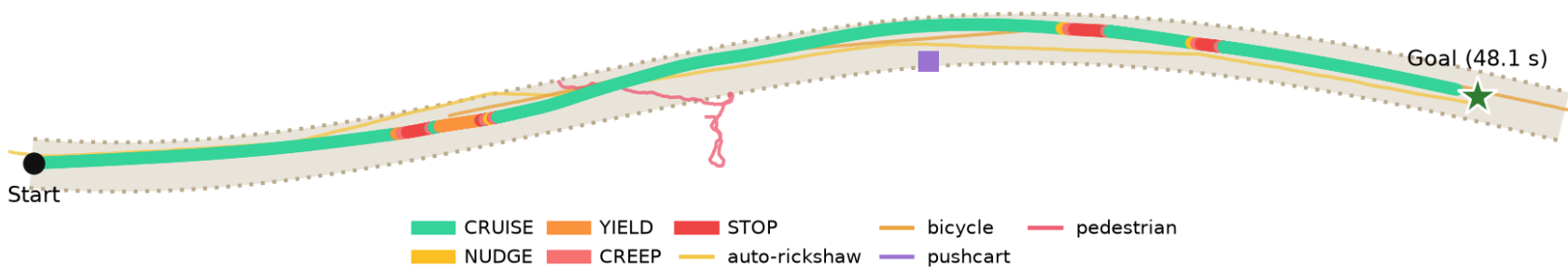


Technologies Used

- MATLAB R2025 / GNU Octave**
Toolbox-free algorithm core
- Simulink + Stateflow**
Closed-loop model, behaviour chart
- RoadRunner + ASAM OpenDRIVE**
Indian road scenes (.xodr)
- YOLO on IDD**
Indian road-user detection demo
- Python · Matplotlib · FFmpeg**
Replays and metrics figures

Implementation Roadmap

- 1 Core loop + village-road scenario ✓
- 2 All 5 scenarios + submission video ✓
- 3 Simulink closed-loop model
- 4 Metrics & technical report, dashboard
- 5 Stateflow, RoadRunner scenes, YOLO-IDD



All 5 scenarios complete collision-free. Shown: the 188 m village road, ego path coloured by behaviour.



FEASIBILITY AND VIABILITY



Technical Feasibility

- **All 5 scenarios run:** sensing → fusion → prediction → planning → control closes end-to-end.
- **Tested:** 8/8 tests pass, including the full closed-loop regression run (EKF, Frenet, Hungarian, collision, bicycle model).
- **15× faster planner** (919 → 61 ms/cycle) from batched candidate checks.
- **One scenario spec** feeds Octave, Simulink and drivingScenario, so nothing is authored twice.

Measured Results · All 5 Scenarios

Scenario	Time	Coll.	Min clear	Replan p95
Village road	48.1 s	0	0.46 m	92 ms
Crossroads	43.5 s	0	3.33 m	107 ms
Highway merge	38.0 s	0	3.89 m	86 ms
Market street	97.1 s	0	0.13 m	98 ms
Cattle crossing	84.2 s	0	1.47 m	98 ms

5 of 5 completed · 0 collisions · clearance scored on ground truth

Resource Feasibility

Tier 0 · GNU Octave – free, unlimited: algorithms, all 5 scenarios, metrics

Tier 1 · MATLAB Online Basic – free: Simulink closed-loop model

Tier 2 · 30-day trial – used last, only for Stateflow, RoadRunner & YOLO-IDD

Hardware: a standard laptop; no GPU needed.

Potential Challenges & Risks → Strategies for Overcoming Them

#	Challenge / Risk	Strategy to Overcome
1	Replan latency crosses the 100 ms budget on busy cycles (p95 86–107 ms by scenario)	Batching already cut cost 15×; next, adaptive lattice size and MATLAB Coder C code
2	Hybrid A* fallback is written and verified, but enabling it made results worse	Kept off: it raised p95 to 250–400 ms and cost a completion; re-enable once search cost and margins are reconciled
3	Dense market clears by only 0.13 m at its tightest	Most margin-sensitive scenario; tune the safety margin and re-validate all five runs
4	COCO-trained YOLO mislabels auto-rickshaws and pushcarts	Class-agnostic tracking fallback; fine-tune on the IDD dataset when a GPU is available
5	RoadRunner is available only in the one-time 30-day trial	Hand-author OpenDRIVE scenes first, so trial days go to assembly, not development



IMPACT AND BENEFITS



1.68 lakh

road deaths in India in 2022 (MoRTH): the problem we target

5

Indian road scenarios — village, crossroads, merge, market, cattle — all completing

0

collisions across all 5 scenarios, scored against ground truth

₹0

licence cost to develop and verify the algorithm core

Who Benefits



Automotive OEMs & Tier-1s

ADAS/AV behaviour tuned for Indian traffic



Logistics & Last-Mile

Autonomous delivery pods, campus shuttles



Rural Mobility

Agri-vehicles and buses on unmarked roads



Students & Researchers

Open, toolbox-free baseline and scenarios



Road-Safety & Test Agencies

Repeatable Indian-road test scenarios

Benefits of the Solution



Social

- Class-aware caution around pedestrians, cyclists & cattle, the most vulnerable road users
- Legible yield/creep behaviour that other road users can anticipate



Environmental

- Jerk-limited, smooth speed profiles avoid harsh braking and wasted energy
- A natural fit for electric autonomous vehicles and low-emission mobility



Economic

- Free-tier development cuts R&D and licence cost
- Reusable scenario library lowers validation cost; indigenous tech for Atmanirbhar Bharat



Technological

- Benchmark scenarios and metrics for unstructured roads
- Stateflow chart keeps every decision explainable to judges and regulators



RESEARCH AND REFERENCES



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Research Insights That Shaped the Design

Lanes are the wrong prior

IDD shows Indian scenes break lane-centric assumptions, so we plan in a free-space corridor [3].

Keep uncertainty visible

Multi-hypothesis prediction feeds a risk field instead of collapsing to one guess.

Frenet lattices give comfort

Jerk-optimal quintic/quartic trajectories are smooth by construction [1].

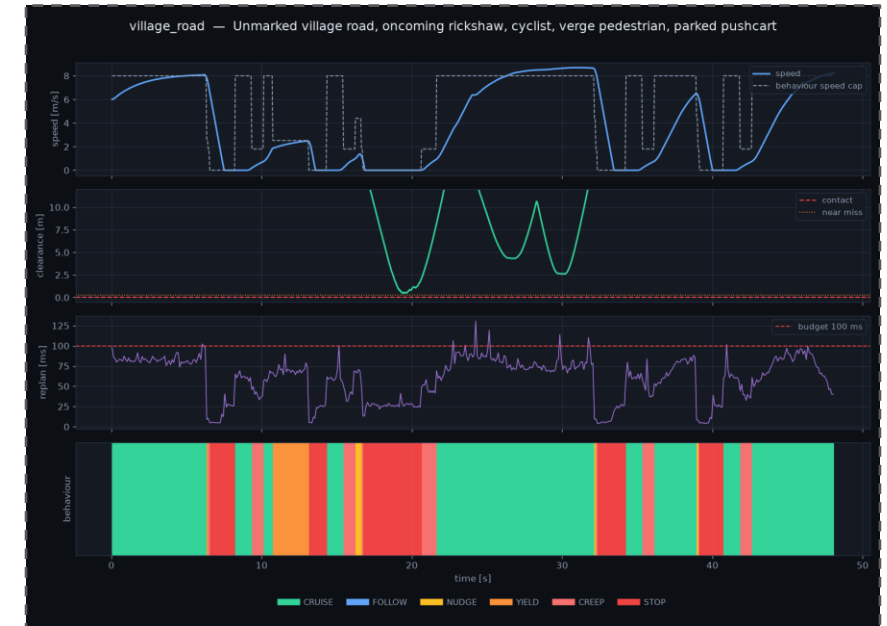
Score safety on ground truth

Clearance measured from our own tracks would be circular, so we use true positions.

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- [3] G. Varma, A. Subramanian, A. Namboodiri, M. Chandraker, C. V. Jawahar, "IDD: A dataset for exploring problems of autonomous navigation in unconstrained environments," IEEE WACV, 2019.
- [4] R. C. Coulter, "Implementation of the pure pursuit path tracking algorithm," CMU-RI-TR-92-01, Carnegie Mellon University, 1992.
- [5] H. W. Kuhn, "The Hungarian method for the assignment problem," Naval Research Logistics Quarterly, vol. 2, 1955.
- [6] S. Thrun, W. Burgard, D. Fox, Probabilistic Robotics, MIT Press, 2005.
- [7] Ministry of Road Transport & Highways, "Road Accidents in India 2022," Government of India, 2023.
- [8] ASAM e.V., "ASAM OpenDRIVE" road-network standard; MathWorks RoadRunner & Automated Driving Toolbox documentation.

Prototype Evidence



Project Links

1) Source Code Repository

github.com/FX-at-Dev/Adaptive-Lane-Free-Path-Planning---Collision-Avoidance-for-Indian-Roads

2) Demo Video: All 5 Scenarios

3) Metrics & Run Data (JSON)